A Benchmark Evaluation of Named Entity Recognition with Large Language Models in Vietnamese

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***Abstract*—We establish the first comprehensive benchmark for Vietnamese Named Entity Recognition (NER) using Large Language Models (LLMs), addressing the gap in low-resource language processing. Our study evaluates leading models—including GPT-5, GPT-4.1, DeepSeek-V3, Qwen3, and Gemma3—across five prompting paradigms ranging from Zero-shot to Tool-Augmented Syntactic Prompting. Testing on VLSP 2018 and COVID-19 NER datasets, we find that task-specific structured prompting (GPT-NER) significantly exceeds the performance of standard prompting techniques. Detailed error analysis reveals critical bottlenecks, notably the semantic overlap between LOC and ORG entities. This work offers a standardized reference and a robust analytical framework for advancing information extraction in linguistically complex, low-resource environments. Our code is available at GitHub[1].**

***Index Terms*—Named entity recognition, domain adaptation, large language models**

I. Introduction

Named Entity Recognition is one of the fundamental tasks in Natural Language Processing (NLP), aiming to identify and categorize meaningful entities in text—such as person names, organizations, locations, dates, or monetary values [11]. NER has become a crucial component in many NLP pipelines, including information extraction, information retrieval, machine translation, and question answering systems [18]. This importance stems from its ability to transform unstructured text into structured entities, thereby enhancing language understanding and knowledge extraction.

Recently, with the rapid advancement of large language models, numerous studies have leveraged their power to improve performance across various NLP tasks, including NER because of their vast knowledge base and strong reasoning capabilities. Works such as [1], [4], [5], [10], [12], [20] have proposed a variety of approaches and benchmarks that effectively integrate LLMs into NER tasks across multiple domains, significantly improving performance. These findings suggest that applying LLMs currently represents one of the most effective and promising directions for solving NER in the present.

However, most existing studies have focused primarily on high-resource languages, while research on low-resource languages—particularly Vietnamese—remains limited. Vietnamese NER studies so far have mainly developed deep learning and pre-trained model-based approaches [7], [13]–[16] which are gradually becoming outdated in the era of rapidly evolving LLMs. Linguistic differences between Vietnamese and high-resource languages also pose challenges for direct method transfer especially given the lack of standardized benchmarks for evaluating LLM-based NER systems in Vietnamese.

Motivated by this research gap and the urgent need for evaluation, we conducted this research and our contributions are threefold:

- We introduce the first standardized benchmark for evaluating LLMs on Vietnamese NER, experimenting with widely used prompt strategies as well as NER-specific prompting techniques on two well-known Vietnamese datasets: VLSP 2018 [23] and COVID-19 NER [24].
- We conduct detailed performance and error analyses, providing insights and identifying potential directions for future research on Vietnamese NER.
- We offer a reference framework that can inspire future benchmark construction for LLM-based NER in other low-resource languages.

II. Related Work

In recent years, numerous studies have explored the Named Entity Recognition task from various perspectives. Some works have proposed novel frameworks for NER, such as [5] which introduced the SaM framework that dynamically selects and merges domain-specific expert models at inference time, and [4] which presented ZeroNER—a description-driven framework designed to enhance NER performance under low-resource conditions. In addition, [12] proposed IFWRANER, a retrieval-augmented large language model fine-tuned through instruction learning for NER, while [6] investigated the impact of data augmentation on confidence calibration and uncertainty estimation in NER. Furthermore, several studies [8], [19]–[21] have developed prompt-based approaches specifically tailored for NER, yielding remarkable improvements over conventional methods. Beyond methodological advances, [1] established comprehensive NER benchmarks, while others [2], [3], [10] introduced new datasets across diverse domains. In addition to high-resource languages, there have also been studies focusing on low-resource languages, such as [17], [22], thereby enriching the landscape of NER research.

In contrast to the diversity found in high-resource languages, research on Named Entity Recognition in Vietnamese remains limited in both scope and quantity. Most existing studies focus primarily on dataset construction; for example, [23]

[1]https://github.com/NERVIETNAMESE

and [24] are considered two widely used benchmark datasets for Vietnamese NER. Recently, several additional datasets have been introduced, such as VietBioNER [15], a manually annotated biomedical NER corpus focusing on tuberculosis treatment, and VietMed-NER [9], the first spoken NER dataset in the Vietnamese medical domain. VietMed-NER is currently the largest real-world Vietnamese NER dataset in terms of entity type coverage, featuring 18 distinct entity types. Beyond dataset creation, some studies have proposed new models or methodologies for Vietnamese NER. For instance, [14] introduced NER2QUES, a method designed to address NER for low-resource languages like Vietnamese. Similarly, [16] proposed a novel fine-tuning approach for multilingual models to handle open-domain NER in Vietnamese. An end-to-end platform for Vietnamese real-estate NER was presented in [13].

From this survey of related work, it is evident that current approaches to Vietnamese NER are outdated and lag behind the rapid advancements of large language models. Therefore, our study aims to establish the first standardized benchmark for evaluating LLMs on Vietnamese NER, providing a solid foundation for future research in this area.

III. Corpora

In this study, we evaluate on two Vietnamese NER test sets.

A. VLSP 2018

The VLSP 2018 [23] dataset is a standard benchmark corpus for Vietnamese NER, consisting primarily of news articles collected from major online newspapers. Table I show some statistic about this test dataset.

Table I: Detailed statistics and entity distribution of the VLSP 2018 test dataset.

Category	Statistic / Entity Type	Value
General Statistics	Total sentences	2,831
	Total tokens	66,097
	Average sentence length	23.35 tokens
Entity Distribution	Number of entity types	4
	Total entities (N)	2,994
	LOC (Location)	45.99%
	MISC (Miscellaneous)	1.64%
	ORG (Organization)	9.15%
	PER (Person)	43.22%

Entity Definitions:

- **PER:** Proper names of individuals or persons.
- **ORG:** Names of organizations, companies, or institutions.
- **LOC:** Geographical locations (countries, cities, seas). Excludes nationalities; may overlap with organizations.
- **MISC:** Names of works, events, or products that include personal or geographical names but do not directly refer to those entities.

B. COVID-19 NER

The COVID-19 NER [24] dataset is the first manually annotated Vietnamese dataset focusing on the COVID-19 domain. It was collected from reputable online news sources in Vietnam. Table II show some statistic about this test dataset.

Table II: Comprehensive statistics and entity distribution of the COVID-19 NER test dataset.

Category	Metric / Entity Type	Value
General Statistics	Total sentences	3,000
	Total tokens	85,678
	Average sentence length	28.56 tokens
Entity Distribution	Number of entity types	10
	Total entities (N)	11,735 (100%)
	LOCATION	37.84%
	PATIENT_ID	17.09%
	DATE	14.09%
	SYMPTOM_AND_DISEASE	9.68%
	ORGANIZATION	6.57%
	AGE	4.96%
	GENDER	3.94%
	NAME	2.71%
	TRANSPORTATION	1.64%
	JOB	1.47%

Entity Definitions:

- **PATIENT_ID:** Unique national-level identifier of a COVID-19 patient. For ranges (e.g., "BN990–993"), each ID is labeled separately.
- **NAME:** Personal name of a patient or a direct contact.
- **AGE:** Numeric age of a patient or contact. For ranges, each number is labeled separately; the word "years" is excluded.
- **GENDER:** Gender information (male/female) linked to an identified individual; excludes honorifics and gendered nouns.
- **JOB:** Occupation of a patient or contact person; labeled only when the field of expertise is specified.
- **LOCATION:** Locations in the movement of patient history (countries, provinces, cities, public places, or venues). For overlaps, label the longest span.
- **ORGANIZATION:** Institutions associated with the patient. Typically, organizations act as agents, while locations denote physical places.
- **SYMPTOM_AND_DISEASE:** Symptoms or pre-existing conditions. For overlaps, label the longest span; exclude pathogens and negated symptoms.
- **TRANSPORTATION:** Transportation identifiers used by the patient. Label IDs only, not vehicle types.
- **DATE:** Dates in formats "X/Y" or "X–Y", including ranges; label each date in a range separately.

IV. Evaluation Methodology

A. Evaluation Metric

In this study, we evaluated our Named Entity Recognition model using micro-averaged precision, recall, and F1-

score with the seqeval library. This library assesses NER performance by treating entities as complete spans in the BIO tagging scheme, requiring exact boundary matches to avoid partial credit. Precision measures the proportion of correctly predicted entities, recall captures how many true entities were identified, and F1-score provides their harmonic mean for balanced evaluation. We adopted micro-averaging to address class imbalance, as it aggregates results over all entity instances, yielding a more stable and realistic performance measure.

B. Evaluation Models

We conduct experiments using a diverse range of widely adopted large language models, including open-source models such as *DeepSeek-V3*, *Qwen3-32B*, *Gemma3-27B-it*, *LLaMA-3.3-Swallow-70B*, and *LLaMA-3.3-70B-Instruct*, as well as closed-source models from the GPT family, namely *GPT-5*, *GPT-4.1*, *GPT-4.1-mini*, *GPT-4o*, and *GPT-4o-mini*. However, due to limited budget, it was not feasible to evaluate all models across every proposed method or dataset.

C. Evaluation Methods and Implementation

To assess the performance of large language models on Vietnamese NER tasks, we evaluate five prompting methods, grouped into two categories: **(1) general prompting strategies** adapted from prior work [17], and **(2) NER-specific prompting methods** designed specifically for entity extraction.

a) General prompting strategies.:

1) **Zero-shot:** This method provides the model with only a task description and entity definitions, without any examples, relying on the pre-trained knowledge of model to perform NER on unseen inputs. In our implementation, the prompt contains the task description, entity definitions, and the input sentence. A single API call returns a structured JSON output, followed by post-processing to parse and clean tags.
2) **Few-shot:** This technique includes a small number of labeled demonstrations to guide the model in understanding entity types and the BIO tagging format for better generalization. We prepend 6 training examples from training set of each datasets to the original zero-shot prompt, allowing the model to learn entity extraction patterns from annotated data while keeping the overall prompt structure consistent with the zero-shot setting.
3) **Chain-of-Thought (CoT):** This strategy elicits intermediate reasoning steps before generating the final NER output, improving complex decision-making. The prompt explicitly introduces reasoning steps and instructs the model to: (1) interpret the task and relevant entity categories; (2) identify and classify candidate entities step by step; (3) remove duplicates to ensure uniqueness; and (4) format the final answer in structured JSON form.

b) NER-specific prompting methods.:

1) **GPT-NER** [20]. This method bridges sequence labeling and text generation by reformulating NER as a marked-text generation task, where entities are wrapped with special tokens (e.g., `@@entity##`). It further introduces a self-verification step to mitigate hallucinations: after generating marked entities, a separate API call validates whether each extracted span is a valid entity of the specified type. The prompt includes the task description, entity definitions, and few-shot examples from the training data. Our implementation uses `extract_entities` for API calls, regular expressions for span extraction, alignment to BIO tags, and post-cleaning.
2) **Tool Augmentation and Syntactic Prompting** [21]. This approach augments the extraction prompt with syntactic information (Part-of-Speech (POS) tags) generated by an external tool and encourages step-by-step syntactic analysis to improve recognition accuracy. POS tags are generated first and then injected into the NER prompt together with reasoning hints. We adopt a decomposition-based QA strategy that processes labels sequentially, and apply self-consistency by sampling multiple predictions and performing majority voting across spans and types. In the codebase, feature flags in `ner_model.py` enable these components, while `predict_tags` handles merging and validation of the final outputs.

V. Result and discussion

A. Overall Performance Comparison

Based on the results presented in Table III, the **GPT-NER** [20] approach achieves the best performance across all three evaluation metrics with two datasets, with **DeepSeek-V3** emerging as the top-performing model (VLSP 2018: precision = 0.72, recall = 0.77, and F1-score = 0.75), (COVID 19: precision = 0.74, recall = 0.85, and F1-score = 0.79). Across all evaluated methods, a clear trend can be observed: models with a larger number of parameters tend to yield higher performance, as exemplified by **DeepSeek-V3** and **GPT-4.1**. Notably, **Gemma3-27B-it** demonstrates consistently superior performance compared to most other open-source large language models of similar or even larger scale (except for **DeepSeek- V3**) across all evaluated methods. When comparing *closed-source* models (GPT family) and *open-source* ones (Gemma, LLaMA, Qwen), the results show that closed-source models still maintain a clear advantage in overall performance. However, the strong results achieved by **DeepSeek-V3** highlight the growing potential of this open-source model that are optimized for Vietnamese NLP tasks. These findings lay the groundwork for more detailed analyses in the following sections, particularly regarding the impact of different prompting strategies.

B. Prompting Strategies Analysis

This section analyzes the strengths and weaknesses of each prompting strategy used in our experiments.
Zero-shot [17].

Table III: Performance comparison across methods, models, and datasets.

Method	Model	VLSP 2018			COVID-19		
		Precision	Recall	F1-score	Precision	Recall	F1-score
Zero-shot	Qwen3-32b	–	–	–	0.47	0.16	0.24
	Llama-3.3-Swallow-70B	0.29	0.37	0.33	0.31	0.28	0.29
	Llama-3.3-70B-Instruct	0.27	0.38	0.32	0.32	0.36	0.34
	Gemma3-27b-it	0.39	0.48	0.43	0.36	0.40	0.38
	GPT-4o-mini	0.31	0.37	0.34	0.22	0.23	0.23
	GPT-4.1-mini	0.57	0.64	0.60	0.47	0.56	0.51
	GPT-4o	0.57	0.36	0.44	0.46	0.56	0.51
	GPT-5	–	–	–	0.47	0.57	0.51
	GPT-4.1	0.55	0.70	0.62	0.51	0.65	0.57
	DeepSeek-V3	0.51	0.72	0.59	0.55	0.72	0.63
Few-shot	Qwen3-32b	–	–	–	0.53	0.17	0.26
	Llama-3.3-Swallow-70B	0.31	0.37	0.34	0.37	0.32	0.34
	Llama-3.3-70B-Instruct	0.27	0.38	0.31	0.32	0.35	0.34
	Gemma3-27b-it	0.33	0.50	0.40	0.36	0.43	0.39
	GPT-4o-mini	0.29	0.40	0.34	0.28	0.31	0.29
	GPT-4o	0.55	0.62	0.58	0.52	0.58	0.55
	GPT-4.1-mini	0.51	0.67	0.58	0.52	0.64	0.58
	GPT-4.1	0.45	0.63	0.53	0.57	0.67	0.62
	DeepSeek-V3	0.61	0.72	0.66	0.64	0.78	0.70
CoT	Qwen3-32b	–	–	–	0.50	0.13	0.21
	Gemma3-27b-it	0.35	0.42	0.38	0.37	0.40	0.38
	GPT-4o-mini	0.09	0.11	0.10	0.17	0.19	0.18
	GPT-4.1-mini	0.44	0.62	0.51	0.41	0.56	0.47
	GPT-4o	0.55	0.62	0.58	0.43	0.51	0.47
	GPT-4.1	0.50	0.68	0.58	0.53	0.64	0.58
	DeepSeek-V3	0.43	0.73	0.54	0.53	0.73	0.62
Tool Aug + SP	Llama-3.3-Swallow-70B	0.26	0.35	0.30	0.32	0.25	0.28
	Llama-3.3-70B-Instruct	0.25	0.36	0.30	0.34	0.35	0.34
	Gemma3-27b-it	0.39	0.42	0.41	0.34	0.38	0.36
	DeepSeek-V3	0.46	0.69	0.55	0.55	0.73	0.63
GPT-NER	Llama-3.3-70B-Instruct	0.70	0.72	0.71	0.63	0.76	0.69
	Llama-3.3-Swallow-70B	0.63	0.63	0.63	0.64	0.78	0.70
	Gemma3-27b-it	0.54	0.76	0.63	0.68	0.84	0.75
	DeepSeek-V3	**0.72**	**0.77**	**0.75**	**0.74**	**0.85**	**0.79**

This strategy is simple and efficient, requiring no example construction and producing responses quickly. However, for low-resource languages such as Vietnamese, the absence of examples and reasoning guidance often leads to suboptimal performance, especially on complex NER tasks.

Few-shot [17].

By incorporating several in-domain Vietnamese NER examples, this strategy generally yields significant performance gains over the zero-shot setting. Its main limitation lies in the need for carefully curated and domain-specific examples, and it still underutilizes the reasoning capabilities of LLMs.

Chain-of-Thought [17].

CoT prompting explicitly guides the model through step-by-step reasoning, effectively leveraging its inference ability. Nevertheless, we observe that long and complex prompts make it difficult for many models to strictly follow output formatting instructions (e.g., returning results in JSON format), which can lead to evaluation errors. Moreover, CoT increases computational cost and inference time due to the extended reasoning process.

GPT-NER [20].

This strategy effectively bridges the gap between sequence labeling and text generation by using special tokens to mark entities, making it well-suited for LLMs. The self-verification step helps reduce hallucinations, leading to more reliable outputs, especially in low- resource settings like Vietnamese NER. However, it requires multiple API calls for generation and verification, which increases latency and computational costs.

Tool Augmentation and Syntactic Prompting [21].

By incorporating POS tags and step-by-step syntactic hints, this strategy enhances the model's reasoning and improves entity recognition on complex sentences. Decomposition by labels and self-consistency via majority voting further boost accuracy and robustness. Nevertheless, generating POS tags adds an extra processing step, and the overall method demands higher resources due to sequential label handling and multiple sampling, potentially limiting its efficiency for large-scale applications.

C. Error Analysis

In this section, we conduct an error analysis based on the predictions of the best-performing method and model, which are **GPT-NER** and **DeepSeek- V3**, respectively, on both datasets. We analyze two types of errors: *false positives*, which refer to cases where the model makes redundant or incorrect

predictions, and *false negatives*, which refer to cases where the model misses or incorrectly predicts entities.

1) VSLP 2018 dataset: Based on the confusion matrix 1, we observed that the **LOC** label has the highest error rate. For *false positive* errors, we found that the model tends to predict words referring to *general areas or regions* as the **LOC** label, whereas **LOC** should only be assigned to specific geographical entities. For example, in the sentence *"Cuộc hành trình như dài ra khi chúng tôi không thể đón taxi đi về miền nam mà phải chuyển sang xe buýt, băng băng đi đến cuối ngày tới Singapore."* (The journey seemed longer when we couldn't take a taxi to the south and had to switch to a bus, traveling all day to reach Singapore), the phrase *"miền nam"* (the south) was incorrectly labeled as **LOC**. This phrase does not refer to a specific place but rather to a general region, which reflects the inherent ambiguity in Vietnamese when distinguishing such expressions — a challenge even for human annotators. For *false negative* errors, the model tends to confuse **LOC** with **PER** when a location name contains a *person's name*, such as streets or schools. For instance, in the sentence *"Đường nối từ đại lộ Nguyễn Văn Linh vào chợ hiện cũng chưa được tráng nhựa, trong khi đường Hoàng Đạo Thuý, con đường ngắn nhất từ khu vực An Lạc để đến chợ, hiện vẫn là một con đường ... làng đắp đất đá với đầy những ổ voi khổng lồ."* (*The road from Nguyen Van Linh Boulevard to the market has not been paved, while Hoang Dao Thuy Street, the shortest route from the An Lac area to the market, remains a dirt village road full of large potholes.*), the model mistakenly identified *"đại lộ Nguyễn Văn Linh"* (*Nguyen Van Linh Boulevard*) as a *person name* (**PER**). This error occurred because the prompt did not provide explicit instructions for how to handle such ambiguous cases, where personal names appear as part of location names.

2) COVID 19 dataset: Based on the confusion matrix 2, we observed that the label with the highest error rate is **ORGANIZATION**. This label is assigned to specific entities such as institutions, agencies, and companies. Regarding *false positives*, we did not observe any such cases, indicating that the model rarely predicts the **ORGANIZATION** label incorrectly when it should not. However, for *false negatives*, our analysis shows that **ORGANIZATION** is most frequently misclassified as **LOCATION**. For instance, in the following sentence: *"Bác sĩ Trần Thanh Linh, từ Bệnh viện Chợ Rẫy chi viện phụ trách đơn nguyên hồi sức tích cực, cho biết bệnh nhân 416 vẫn đang duy trì ECMO, thở máy, hiện xơ phổi rất nhiều."* (*Dr. Tran Thanh Linh from Cho Ray Hospital, who was dispatched to lead the Intensive Care Unit, stated that Patient 416 remains on ECMO and ventilators, and is currently suffering from severe pulmonary fibrosis.*), the phrase *"Bệnh viện Chợ Rẫy"* (*Cho Ray Hospital*) was predicted as a **LOCATION**, whereas it should have been labeled as an **ORGANIZATION**. The distinction here lies in the fact that a hospital represents an institutional organization with personnel, structure, and specific regulations, while **LOCATION** merely denotes a physical place without organizational activity. This type of confusion is inherently challenging, as even humans may misinterpret or use these categories interchangeably in daily communication.

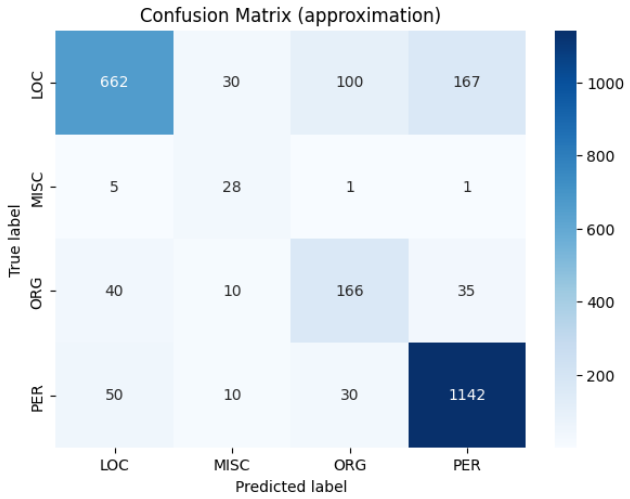


Figure 1: Confusion matrix of GPT-NER with DeepSeek-V3 on the VLSP 2018 dataset.

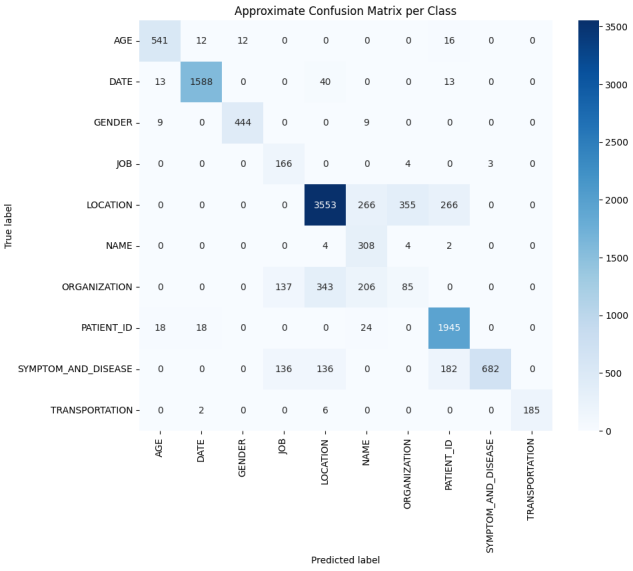


Figure 2: Confusion matrix of GPT-NER with DeepSeek-V3 on the COVID-19 dataset.

Consequently, distinguishing between **ORGANIZATION** and **LOCATION** poses a significant challenge for NER models. The model's errors in this regard are likely due to insufficient training data that clearly differentiates between the two labels, as well as a lack of explicit guidance in the prompt to handle such borderline cases.

D. Challenges and Proposed Solutions

For Vietnamese Named Entity Recognition, based on the previous error analysis, we observe that several major challenges remain. First, some labels contain many generic or ambiguous words that frequently co-occur with tokens belonging to specific entity types, leading the model to produce false or redundant predictions. Second, certain label pairs are

inherently similar and easily confused, even by humans in daily communication. These labels are often distinguished only by subtle contextual or semantic cues, making it difficult for the model to infer the correct classification.

From these observations, we propose several strategies to improve the performance of Vietnamese NER systems. Training or fine-tuning datasets should include detailed explanations and explicit guidance for handling cases where entity types are hard to distinguish. Moreover, prompt-based approaches should incorporate well-crafted, illustrative examples that demonstrate clear reasoning in challenging contexts. Finally, applying example selection or data filtering methods to prioritize high-quality, informative samples rather than random selection may further enhance model robustness and accuracy.

VI. Conclusion & Future Works

This study establishes the first standardized benchmark for evaluating Large Language Models on Vietnamese Named Entity Recognition, identifying the GPT-NER strategy and the open-source DeepSeek-V3 model as top performers among a diverse suite of approaches. Through rigorous experimentation on the VLSP 2018 and COVID-19 datasets, we highlighted persistent challenges in low-resource settings, particularly the semantic ambiguity between Organization and Location entities that necessitates more explicit reasoning guidance.

In future work, we plan to expand this benchmark to include a broader range of both open-source and closed-source LLMs, as well as experiment with more diverse prompting strategies methods and more implementation about fine-tuning LLMs. Furthermore, we aim to develop new techniques to enhance the performance of LLMs on Vietnamese NER, building upon the limitations and insights identified in this study.

VII. Limitation

In this study, due to limited computational and financial resources, we were only able to experiment with a subset of both open-source and closed-source LLMs. Additionally, we have so far evaluated only two methods specifically designed for the NER task.

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